Building a Self-Learning Eye Gaze Model from User Interaction Data

Michael Xuelin Huang, Tiffany C.K. Kwok, Grace Ngai, Hong Va Leong, Stephen C.F. Chan
Department of Computing
Hong Kong Polytechnic University
Hong Kong, China
{csxhuang,csgngai,cshleong,csschan}@comp.polyu.edu.hk, tiffany.ck.kwok@gmail.com

ABSTRACT
Most eye gaze estimation systems rely on explicit calibration, which is inconvenient to the user, limits the amount of possible training data and consequently the performance. Since there is likely a strong correlation between gaze and interaction cues, such as cursor and caret locations, a supervised learning algorithm can learn the complex mapping between gaze features and the gaze point by training on incremental data collected implicitly from normal computer interactions. We develop a set of robust geometric gaze features and a corresponding data validation mechanism that identifies good training data from noisy interaction-informed data collected in real-use scenarios. Based on a study of gaze movement patterns, we apply behavior-informed validation to extract gaze features that correspond with the interaction cue, and data-driven validation provides another level of crosschecking using previous good data. Experimental evaluation shows that the proposed method achieves an average error of 4.06°, and demonstrates the effectiveness of the proposed gaze estimation method and corresponding validation mechanism.

Categories and Subject Descriptors
H.1.2 [**Models and Principles**]: User/Machine Systems—*Human factors*; I.5.m [**Pattern Recognition**]: Miscellaneous.

General Terms
Experimentation; Human Factors.

Keywords
Gaze estimation; supervised learning; implicit modeling; data validation

1. INTRODUCTION
Gaze information, as a reflection of human attention, cognition and even desire and emotion, has great potential applications in a large number of domains, including interaction and diagnostics [7]. As a result, eye gaze research has been continuously popular for more than 30 years [8].

In general, gaze estimation methods can be categorized as feature-based or appearance-based [8]. Feature-based methods extract eye features such as eye contour, corners, pupil center and glint(s) (under active light sources), and are able to achieve relatively high accuracy. However, accurate and robust feature extraction requires specialized equipment such as infrared (IR) light source(s) or head mounted devices. Appearance-based approaches use the image itself as input and map it to screen coordinates. These approaches implicitly estimate the relevant gaze features, such as eye geometric features, camera calibration parameters and personal variance. They are more versatile for general use as they do not require any special devices, and they can be interesting for multimedia domains, like quality assessment and gaming using gaze information. However, they are sensitive to noise caused by variance of lighting condition, head pose, and individuality.

There have been some efforts into the use of low-cost off-the-shelf equipment for gaze estimation. Such approaches clearly require the use of gaze features that are robust to visual variances, and an accurate method of obtaining calibration data. Some researchers have proposed to use visual salient information for implicit gaze calibration [1][9], but the computed saliency model is not always consistent with the point of regard (PoR), or the true location of the gaze point on the screen [9]. Lu et al. [12] investigate regression on sparse data and introduce a synthesis framework [11] to produce training samples for unseen head poses. These approaches require explicit calibration and are sensitive to lighting variance. Sugano et al. proposed an alternate model, which collects mouse click data as the ground truth to incrementally update the gaze model [15]. Their approach assumes that users are looking at the point that they click on. However, in unconstrained real-use situations, this may not always be an accurate assumption.

This paper proposes to utilize the numerous interaction-informed gaze points that are generated during regular, daily human computer interaction to train an eye gaze model. These gaze points correspond to interactions with the mouse and keyboard, and can be obtained automatically from the interaction events. Unlike previous work, we take into consideration the *characteristics* of the eye movement patterns during the interaction event to inform our model and identify reliable instances for training. By taking into account both gaze and head pose features, it may be possible to achieve head pose invariance even with standard low-cost equipment. Our objective is to unobtrusively collect reliable training data to build a reliable model for off-the-shelf equipment that fits the individual gaze attributes, such as inter-ocular distance, eye and face contour, etc.

To the best of our knowledge, there is no prior work that investigates automated data validation techniques for gaze model learning from daily, noisy interaction cues. The contributions of this paper are thus as follows: we propose (1) an interaction-informed method for practical collection of data for gaze modeling, (2) a set of robust features for gaze estimation, and (3) the corresponding data processing and validation mechanism to identify good interaction-informed data for training.

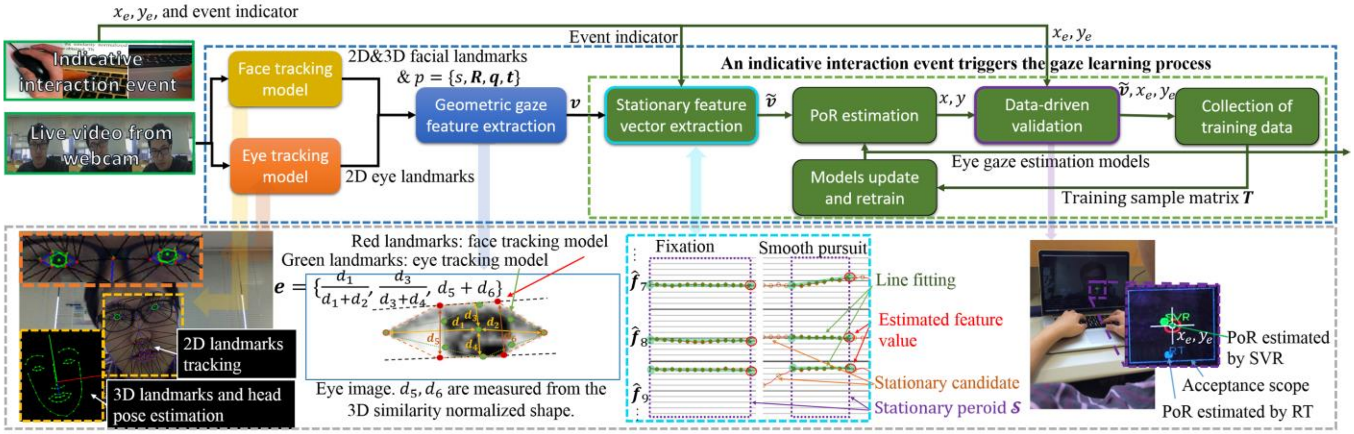

Figure 1. System Schematic

2. METHODOLOGY

Figure 1 shows our system schematic. A standard webcam streams a live video of the user's head and shoulders while his mouse movements and keystrokes are tracked by the system. Two independent tracking models are applied to the video stream to extract face and eye landmarks and the corresponding parameters, based on which the gaze feature vector $\widetilde{\boldsymbol{v}}$ is extracted and processed from the eye image.

We start by collecting 100 instances for our initial model. Each instance is collected upon an interaction event; we use the video frame immediately preceding the interaction event to generate our feature vector for training and the location of the mouse click or the caret (for keyboard events) is taken as the PoR of the user. Two machine-learning algorithms, support vector regression (SVR) [6] and regression tree (RT) [10], are used to build PoR estimation models from these initial instances.

Upon the arrival of new interaction events, the estimated PoRs (x, y) from the models are compared with the interaction-informed PoR (x_e, y_e) from the mouse and keyboard data. If the PoRs match, the gaze feature vector $\widetilde{\boldsymbol{v}}$ and the interaction-informed PoR (x_e, y_e) will be retained and used to adapt and retrain the gaze regression model for the user.

2.1 Extracting Gaze Features from Video

2.1.1 Head Pose Estimation

To obtain accurate locations of the facial landmarks and head pose information, we apply Saragih's Constrained Local Model (CLM) [14]. CLMs generate parametric models for face alignment based on the point distribution model (PDM) [5] and the localized patch model, which reflect the interdependency and the local appearance of the landmarks, respectively.

We use a 3D CLM model in our experiments. The 3D vertices of the facial landmarks can be represented by $\boldsymbol{x} = (\boldsymbol{x}_1, \dots, \boldsymbol{x}_M)^T$, where $\boldsymbol{x}_i = (x_i, y_i, z_i)$ denotes the coordinate of the i^{th} landmark in the camera coordinate and M(=66) is the number of facial landmarks (see notation[1]). Procrustes alignment [4] is used to normalize the rigid transformation and Principal Component Analysis (PCA) is applied to approximate the non-rigid deformation of the given samples. Therefore, the face images and corresponding annotated landmarks can be represented by $\boldsymbol{x}_i(p) = s\boldsymbol{R}(\bar{\boldsymbol{x}}_i + \boldsymbol{\Phi}_i\boldsymbol{q}) + \boldsymbol{t}, i \in [1, M]$, where $\boldsymbol{x}_i(p)$ and $\bar{\boldsymbol{x}}_i$ represent the 3D and mean coordinates of the i^{th} landmark, respectively; $\boldsymbol{\Phi}_i$ the sub-matrix of the basis of variation of the i^{th} landmark; and $\boldsymbol{p} = \{s, \boldsymbol{R}, \boldsymbol{q}, \boldsymbol{t}\}$ the PDM parameters: a head scaling s, a three-axis orientation $\boldsymbol{R}$, a global head translation $\boldsymbol{t}$ and a non-rigid transformation vector $\boldsymbol{q}$.

Generally, the objective of CLM fitting is, given an image $\mathcal{I}$, to determine the PDM parameter $\boldsymbol{p}$ so as to minimize the misalignment of the landmark with the cost function $f(\boldsymbol{p}) = \mathcal{R}(\boldsymbol{p}) + \sum_{i=1}^{M} \mathcal{L}_i(\boldsymbol{x}_i; \mathcal{I})$. This can also be viewed as maximizing the likelihood of both the global geometric shape deformation $\mathcal{R}(\boldsymbol{p})$ and the local patch textural information of the landmarks, $\sum_{i=1}^{M} \mathcal{L}_i(\boldsymbol{x}_i; \mathcal{I})$. However, due to the nature of the training image data and the influence of local maxima in optimization, CLM tracking tends to fail under poor lighting condition and certain kinds of expressions, e.g. mouth corner depression. To improve localization accuracy of the landmarks and to facilitate head pose estimation, we use the Supervised Descent Method (SDM) to optimize 48 facial landmarks [16]. Unlike CLM, SDM learns the sequence of descent directions for individual landmarks from training data, by minimizing the following function over the descent direction $\Delta\boldsymbol{x}$, $f(\boldsymbol{x}_c + \Delta\boldsymbol{x}) = \|\mathcal{A}(\boldsymbol{x}_c + \Delta\boldsymbol{x}) - \mathcal{A}_*\|_2^2$, where $\boldsymbol{x}_c$ denotes the initial configuration of the landmarks, $\mathcal{A}$ represents the appearance feature extraction and $\mathcal{A}_*$ the appearance feature near the manually annotated landmarks. During tracking, SDM optimizes the landmark localization with the learnt descent directions.

Our approach uses SDM for fast and robust face tracking. For each frame, we use the SDM fitting results as the initial state for CLM fitting. This improves the localization accuracy and reduces the time needed for convergence. CLM optimization then provides us with the 2D and 3D facial landmarks and the parameters $\boldsymbol{p}$. By removing the rigid geometric variation ($s, \boldsymbol{R}$ and $\boldsymbol{t}$), the similarity-normalized shape, i.e. the aligned 3D landmarks, can be obtained.

2.1.2 Pupil Center Estimation

An accurate estimation of gaze direction requires knowing the location of pupil center. However, in unconstrained situations, the appearance information of the eye region usually fails to provide a clear cue for the pupil center. Additionally, we observe that quite often in real-use scenarios, low video resolution and reflections on glasses and cornea may make the region of the pupil and its periphery unobservable.

[1] Bold capital letter denotes a matrix $\boldsymbol{F}$, bold lower-case letters a column vector $\boldsymbol{f}$. $\boldsymbol{f}_j$ represents the j-th column of the matrix $\boldsymbol{F}$. f_{ij} denotes the scalar in the row i and column j of the matrix $\boldsymbol{F}$.

To address issues caused by the unconstrained environment and noisy images with low resolution, we use eye geometric features to estimate the pupil center. The general idea is to track landmarks with good salient features on the iris contour and eye lid corners, and deduce the pupil center locations based on their geometric interdependency. We therefore apply eye CLMs to track the eye landmarks, including the pupil center (Figure 1). To validate the eye tracking state, we apply a robust but rough tracking model using the integral image [13] to approximate the pupil center from the eye image, with the assumption that the darkest rectangular area is the iris. If the CLM estimation is too distant from the integral image approximation, we assume that the CLM tracking has failed and we redetect the eye for tracking.

Based on the landmarks identified from the face and the eye trackers, we extract n (=12) gaze features and construct a gaze feature vector $\boldsymbol{v} = [s, \boldsymbol{R}, \boldsymbol{t}, \boldsymbol{e}_r, \boldsymbol{e}_l]^T$, $\boldsymbol{v} \in \Re^{n\times 1}$, where $s, \boldsymbol{R}, \boldsymbol{t}$ are the head pose features obtained from CLM, while $\boldsymbol{e}_r, \boldsymbol{e}_l$ denote the eye features from both eyes. Figure 1 shows the eye features, which include eye rotations (yaw and roll) and eyelid openness.

2.2 A Two-Layer Mechanism for Validating Collected Data

The most frequent gaze patterns occurring during human computer interaction are fixations, smooth pursuits, saccades and blinks. Fixation indicates a stationary gaze. Smooth pursuit denotes relatively slow gaze movements, while saccades refer to eye movements that rapidly direct towards a stationary target. Sugano et al. [15] collected data from mouseclicks based on the assumption that people tend to fixate on what they click. Our observations, however, suggest that in normal human-computer interactions, this assumption that the position of the mouseclick is the PoR at that time instance may not be valid. By the time the click occurs, the gaze may have already moved, or the gaze may be on a large image link. In addition, mouse interactions represent only a small percentage of human interactions with the computer.

We generalize Sugano et al's assumption to other interaction activities, and we postulate that people's gaze patterns are highly consistent with particular interaction events. Table 1 shows some examples. In addition, we focus on identifying periods of fixation and smooth pursuit as time periods where the PoR of the user is most likely to correlate with the location of the interaction event.

2.2.1 Behavior-Informed Validation – Extracting a Stationary Feature Vector

We refer to the gaze feature during fixation and smooth pursuit as "stationary" or "stationary with small trend" for simplicity. Since the raw gaze feature vector is extracted based on the identified landmarks from the statistical tracking models, it contains high frequency noise caused by temporal jitter from landmark localization. For each feature, we therefore perform a linear interpolation to resample the feature signal (sample rate: 15Hz) over the previous 1-second window prior to the interaction event. When an event is captured, we use the feature vectors in the previous m(=15) frames prior to the event to construct the gaze feature matrix $\boldsymbol{F} \in \Re^{m\times n}$. In addition, low-pass filtering in the frequency domain is applied to remove the high frequency temporal jitter from the signal while maintaining the main component without overmuch distortion. The filtered signal is calculated as $\hat{\boldsymbol{f}}_j = \mathcal{F}^{-1}[\mathcal{F}(\boldsymbol{f}_j) * \mathcal{H}(\omega)]$, where $\mathcal{F}$ and $\mathcal{F}^{-1}$ indicate the one-dimensional forward and inverse Discrete Fourier transform, respectively, and the filter $\mathcal{H}(\omega) = \begin{cases} 1, & \omega \le \omega_h \\ 0, & \text{otherwise} \end{cases}$, where ω_h (=2.5Hz in our work) is the cutoff frequency.

Table 1. Examples of Interaction-Informed Gaze Patterns

Interaction event	Human intention	Potential gaze pattern and informed PoR
Mouse left button click	Link or button selection	Fixation/smooth pursuit on the mouse cursor
Mouse left button double-click	Word selection in document editing	Fixation on the mouse cursor
Keyboard button down	Word typing	Fixation/smooth pursuit on the typing caret

We then determine when the signals are in a stationary state. We apply a threshold $\boldsymbol{\varepsilon} \in \Re^{n\times 1}$ to evaluate the temporal variations of the features. $\boldsymbol{\varepsilon}$ is calculated as $\boldsymbol{\varepsilon} = 0.05\boldsymbol{r}$, where $\boldsymbol{r} \in \Re^{n\times 1}$ is the range of the feature values. The frame whose corresponding feature vector satisfies the following condition is considered to potentially be within the stationary period,

$$\prod_{j=1}^{n} H(\varepsilon_j{}^2 - \dot{\hat{f}}_{ij}^2) = 1 \quad (1)$$

where $H(x) = (1 + \mathrm{sgn}(x))/2$ is the Heaviside step function and $\dot{\hat{f}}_{ij}$ is the value of the i^{th} frame of the 1st order signal of $\hat{\boldsymbol{f}}_j$. After all m frames in the feature matrix $\boldsymbol{F}$ have been tested, a backward search is performed to locate the stationary period that is closest to the interaction event and has a duration longer than 0.3 second (3 times the minimum fixation duration [8]). This resulting period, represented as a sequence of frames $\mathcal{S}$, is considered the corresponding stationary state to the interaction event.

Since features in the stationary state are relatively stable, we apply linear regression to approximate the final gaze feature vector $\tilde{\boldsymbol{v}}$. M-estimator is used to iteratively fit the line using the weighted least-squares algorithm to minimize $\sum_{i\in\mathcal{S}} \rho(d_{ij})$, where $\rho(d) = 2(\sqrt{1 + d^2/2} - 1)$ is the distance function and d_{ij} denotes the difference between $\hat{f}_{ij}$ and its fitted value $\tilde{f}_{ij}$. Setting k as the last frame of $\mathcal{S}$, the resulting value for the j^{th} gaze feature should be $\tilde{f}_{kj}$. This generates a final input feature vector for the classifier, $\tilde{\boldsymbol{v}} = [\tilde{f}_{k1}, \dots, \tilde{f}_{kn}]^T$.

2.2.2 Data-Driven Validation

The behavior-informed validation looks for the stationary period corresponding to the received interaction event. However, it is possible that for some interactions, the user's PoR may not be anywhere near the location of the interaction event – for example, when the user is watching a movie with the mouse pointer poised over the "pause" button. The data-driven validation uses the previous collected data as an additional point of validation to determine the goodness of the feature vector $\tilde{\boldsymbol{v}}$ and the corresponding assumed interaction PoR (x_e, y_e).

Passing $\tilde{\boldsymbol{v}}$ through the trained SVR/RT models generates estimated PoRs (x_{SVR}, y_{SVR}) and (x_{RT}, y_{RT}). Comparing the estimated PoRs with the location of the interaction event, (x_e, y_e) gives us estimation errors e_{SVR} and e_{RT}. The feature vector $\tilde{\boldsymbol{v}}$ and the corresponding interaction-informed PoR (x_e, y_e) is retained if $\boldsymbol{min}(e_{SVR,}e_{RT}) \le \lambda$, where λ is a threshold that is set to 1/10 of the screen diagonal in our work. The validated gaze vector and its interaction-informed PoR (x_e, y_e) is stored into the training sample matrices ($\boldsymbol{T} \in \Re^{14\times 100w}$, w is the number of matrices), of which the i^{th} column $\boldsymbol{t}_i = [\tilde{\boldsymbol{v}}_i{}^T, x_{ei}, y_{ei}]^T$. $\boldsymbol{T}$ is used to retrain and update the SVR/RT after every 100 new instances collected.

Our approach therefore uses a dual-level validation. First of all, by requiring the gaze features to be stationary, the behavior-informed

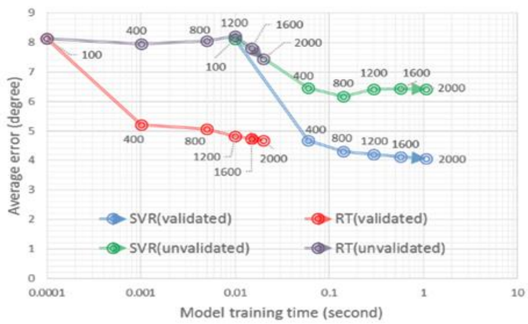

Figure 2. Performance and training time comparison between SVR and RT with/without the data validation mechanism

validation ensures the interaction-informed correspondence through the gaze movement patterns. Secondly, the data-driven validation uses the current gaze estimation model, trained on previously-collected data, and the location of the interaction event, to filter out potentially noisy input data.

3. EXPERIMENTAL EVALUATION

To demonstrate the effectiveness of the proposed method for gaze estimation and the corresponding data validation mechanism, we recruited 5 subjects (3 female, aged 20-30). They were free to adjust head pose, body posture and chair height for their comfort during the experiments. All the experiments were run on a standard 15-inch MacBook Pro, with the screen resolution set to 1920x1200, and the built-in webcam used for data collection. Approximated by the CLM results, the head to screen distance in the training data ranges from 305~634mm (mean: 533mm; sd: 59mm) and the head pitch from -6 º~24 º(mean: 12.6 º; sd: 5.5 º).

Subjects were instructed to work on tasks that involved interactions, such as writing and browsing. (This being the end of the semester, most subjects chose to work on their term reports.) The experiment finished once sufficient keyboard and mouse events (2,000 instances) were collected, approximately 3 hours on average. This is longer than the 10-minute data collection process from previous work [15], but it should be noted that (1) our data collection is implicit rather than explicit, and (2) the previous work does not address the issue of inconsistency between PoR and interaction event location.

For testing, we followed Cerrolaza et al's protocol [3] to capture gaze on 8x8 grids at different distances (450, 500, 550 mm in our case). Two state-of-the-art regression models are evaluated (SVR and RT) in the experiments. Radial basis function kernel is applied for SVR as suggested by Bennett et al. [2].

Figure 2 shows the relation between the model training time and the performance of the gaze estimation models. The performance is measured by the average error [15] over all calibration points and subjects. For comparison, we present the performance on both validated and unvalidated data. The red and blue markers indicate the results of models trained with validated data, while the purple and green represent models that use all interaction-informed data. Different markers denote the performance of increasing amounts of training data, at 100, 400, 800, 1200, 1600 and 2000 instances. The models trained on validated data achieve average error of 4.06 º for SVR and 4.68 º for RT (i.e. mean absolute error: 37 and 43 mm respectively). This is comparable to state of the art work [11][12][15] with similar head motions.

More importantly, by increasing the amount of training data, the error of the validated models decreases smoothly. However, the error of the unvalidated models fluctuate. Moreover, with just 400 training instances, the validated models significantly outperform the best performances achieved on unvalidated data (4.66 º vs. 7.44 º for SVR; 5.20 º vs. 6.16 º for RT). This is because additional data in real-use situations contains much noise, which hurts performance. It corroborates our hypothesis that validation is necessary when working with interaction-informed data collected from real-use situations. This indicates both the effectiveness and importance of our proposed validation mechanism.

4. CONCLUSIONS

This paper proposes a novel approach to extract robust gaze features and the corresponding data validation mechanism to facilitate the use of interaction-informed data for gaze estimation. Experimental evaluations demonstrate that the proposed method can effectively apply daily interaction cues to train an eye gaze model without an explicit calibration phase, achieving performance comparable to that of state-of-the-art methods, but using low-cost off-the-shelf devices. In our future work, we plan to further investigate the impact of data skew and significant head pose variance in long-term experiments.

This work was partially supported by grants PolyU 5248/10E and PolyU 5222/13E from the Hong Kong Polytechnic University and the Hong Kong Research Grants Council, respectively.

5. REFERENCES

[1] Alnajar, F., Gevers, T., Valenti, R. and Ghebreab, S. Calibration-Free Gaze Estimation Using Human Gaze Patterns. In *IEEE ICCV*, 2013, 137-144.

[2] Bennett, K.P. Nonlinear Eye Gaze Mapping Function Estimation via Support Vector Regression. In *ICPR*, 2006, 1132-1135.

[3] Cerrolaza, J.J., Villanueva, A. and Cabeza, R. Study of Polynomial Mapping Functions in Video-Oculography Eye Trackers. *ACM Trans. on Computer-Human Interaction*. 19, 2, 2012, 1-25.

[4] Cootes, T.F., Edwards, G.J. and Taylor, C.J. Active appearance models. *IEEE Trans on Pattern Analysis and Machine Intelligence*. 23, 6, 2001, 681-685.

[5] Cootes, T.F. and Taylor, C.J. Active Shape Models - "Smart Snakes." *BMVC*,1992, 266-275.

[6] Drucker, H., Burges, C.J.C., Kaufman, L., Smola, A. and Vapnik, V. Support vector regression machines. In *NIPS,* 1997, 155-161.

[7] Duchowski, A.T. *Eye Tracking Methodology: Theory and Practice*. Springer London. 2003.

[8] Hansen, D.W. and Ji, Q. In the eye of the beholder: a survey of models for eyes and gaze. *IEEE Trans on Pattern Analysis and Machine Intelligence*. 32, 3, 2010, 478-500.

[9] Judd, T., Ehinger, K., Durand, F. and Torralba, A. Learning to predict where humans look. *IEEE ICCV,* 2009, 2106-2113.

[10] Leo Breiman 2001. Random Forests. *Machine Learning*. 45, 1. 2001, 5-32.

[11] Lu, F., Okabe, T., Sugano, Y. and Sato, Y. Learning gaze biases with head motion for head pose-free gaze estimation. *Image and Vision Computing*. 32, 3, 2014, 169-179.

[12] Lu, F., Sugano, Y., Okabe, T. and Sato, Y. Inferring Human Gaze from Appearance via Adaptive Linear Regression. *IEEE Trans on Pattern Analysis and Machine Intelligence*. 2014, 1-1.

[13] Paul Viola, M.J. Robust Real-time Object Detection. *Int. J. Computer Vision*. 2001.

[14] Saragih, J.M., Lucey, S. and Cohn, J.F. Face alignment through subspace constrained mean-shifts. *ICCV*. 2009, 1034-1041.

[15] Sugano, Y., Matsushita, Y., Sato, Y. and Koike, H. An Incremental Learning Method for Unconstrained Gaze Estimation. In *ECCV,* 2008, 656–667.

[16] Xiong, X. and De la Torre, F. Supervised Descent Method and Its Applications to Face Alignment. *CVPR,* 2013, 532–539.